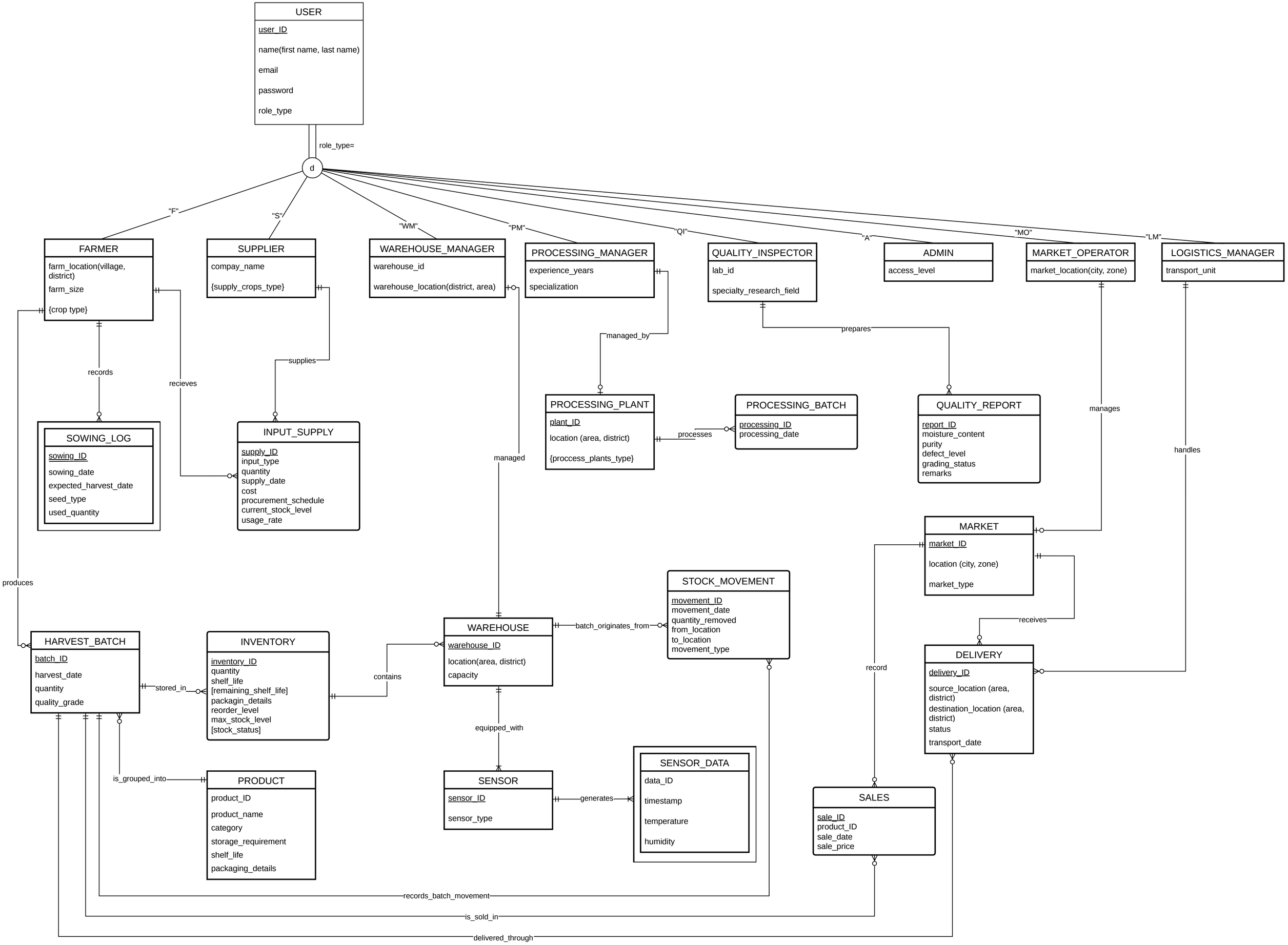
USER
user_ID
name(first name, last name)
email
password
role_type
role_type=
d
"F"
"S"
"WM"
"PM"
"QI"
"A"
"MO"
"LM"
FARMER
farm_location(village, district)
farm_size
{crop type}
SUPPLIER
compay_name
{supply_crops_type}
WAREHOUSE_MANAGER
warehouse_id
warehouse_location(district, area)
PROCESSING_MANAGER
experience_years
specialization
QUALITY_INSPECTOR
lab_id
specialty_research_field
ADMIN
access_level
MARKET_OPERATOR
market_location(city, zone)
LOGISTICS_MANAGER
transport_unit
records
recieves
supplies
managed_by
prepares
manages
handles
SOWING_LOG
sowing_ID
sowing_date
expected_harvest_date
seed_type
used_quantity
INPUT_SUPPLY
supply_ID
input_type
quantity
supply_date
cost
procurement_schedule
current_stock_level
usage_rate
PROCESSING_PLANT
plant_ID
location (area, district)
{proccess_plants_type}
processes
PROCESSING_BATCH
processing_ID
processing_date
QUALITY_REPORT
report_ID
moisture_content
purity
defect_level
grading_status
remarks
managed
MARKET
market_ID
location (city, zone)
market_type
STOCK_MOVEMENT
movement_ID
movement_date
quantity_removed
from_location
to_location
movement_type
produces
receives
HARVEST_BATCH
batch_ID
harvest_date
quantity
quality_grade
stored_in
INVENTORY
inventory_ID
quantity
shelf_life
[remaining_shelf_life]
packagin_details
reorder_level
max_stock_level
[stock_status]
contains
WAREHOUSE
warehouse_ID
location(area, district)
capacity
batch_originates_from
record
DELIVERY
delivery_ID
source_location (area, district)
destination_location (area, district)
status
transport_date
equipped_with
is_grouped_into
PRODUCT
product_ID
product_name
category
storage_requirement
shelf_life
packaging_details
SENSOR
sensor_ID
sensor_type
generates
SENSOR_DATA
data_ID
timestamp
temperature
humidity
SALES
sale_ID
product_ID
sale_date
sale_price
records_batch_movement
is_sold_in
delivered_through

Business Rules

1. Every user in the system must have one unique user_ID and must belong to exactly one role type: Farmer, Supplier, Warehouse Manager, Processing Manager, Quality Inspector, Admin, Market Operator, or Logistics Manager.
2. 2.A Farmer may record many sowing logs, but each sowing log must be recorded by exactly one Farmer. Each sowing log must contain sowing date, expected harvest date, seed type, and used quantity.'
3. A Supplier may provide many input supply records, and a Farmer may receive many input supply records, but each input supply record must refer to exactly one Supplier and one Farmer.
4. A Farmer may produce many harvest batches, but each harvest batch must be produced by exactly one Farmer. Each harvest batch must store harvest date, quantity, and quality grade.
5. A Product may be associated with many harvest batches, but each harvest batch must belong to exactly one Product. Product records must store product name, category, storage requirement, shelf life, and packaging details.
6. A Warehouse may contain many inventory records, but each inventory record must belong to exactly one Warehouse and must reference exactly one harvest batch. Inventory records must store quantity, shelf life, packaging details, reorder level, and max stock level.
7. A Warehouse may be equipped with many sensors, but each sensor must be assigned to exactly one Warehouse. Each sensor may generate many sensor data records, but each sensor data record must come from exactly one sensor.
8. A Processing Plant may process many processing batches, but each processing batch must be processed in exactly one Processing Plant. A Processing Manager is responsible for managing processing plant operations, and a Quality Inspector may prepare many quality reports, but each quality report must be prepared by exactly one Quality Inspector.
9. A Harvest Batch may have many stock movement records and many delivery records, but each stock movement and each delivery must be linked to exactly one Harvest Batch. A stock movement must record movement date, quantity removed, source location, destination location, and movement type.
10. A Market may record many sales, but each sales record must belong to exactly one Market and must refer to exactly one Harvest Batch. A Logistics Manager may handle many deliveries, but each delivery must be handled by exactly one Logistics Manager, ensuring traceability of harvested products from batch to market sale.